

Howard Park

(909) 827-5738 | howard.park@nyu.edu | linkedin.com/in/howardjaypark | hjp309.github.io | Rowland Heights, CA

PROFESSIONAL SUMMARY

Unity developer with seven years of experience building games and the systems around them. Currently sole engineer on WishSeekers, a 2.5D action roguelite: a ScriptableObject data layer routing all damage through one resolution pipeline shared by 30+ abilities, four NavMesh enemy archetypes and a multi-phase boss, and a game-feel stack covering combos, i-frames, parry and input buffering. Earlier work spans shader and performance optimization for a 100k+ user product at Meta, networked multiplayer on Unity Mirror, account and authentication layers on AWS, and integrating a Unity stack into a production Docker pipeline.

CORE COMPETENCIES

Gameplay Systems Architecture

Combat & Enemy AI

Unity 6 & C#

Animation Systems & IK

Multiplayer Netcode

URP/HLSL Shaders

Build & Deployment

Identity & Auth

WORK EXPERIENCE

Meta

December 2023 - December 2024

Technical Artist

- Improved performance for Horizon Worlds (100k+ users) on Unity by writing shaders and optimizing in-house asset creation and profiling tools.
- Wrote a Unity sandbox tool for sound engineers to test ambient and avatar sounds, optimizing noise profiles in high- and low-traffic settings.
- Worked cross-functionally with engineers, artists, and designers, scoping technical feasibility and requirements for new seasonal content.

FiveMics

August 2022 - November 2023

Backend Engineer

- Architected the game's AWS backend and built its account layer end to end: registration, authentication, and cross-stack user-data exchange between the Unity client and native iOS.
- Developed secure REST APIs on API Gateway with Lambda functions to handle account registration, authentication, and user profile management.
- Developed a full-stack onboarding and registration flow with UX artists and designers, supporting Desktop, Google and Apple sign-in via Cognito, DynamoDB and Unity.
- Employed the AWS CLI to deploy Unity builds on CloudFormation for scalable infrastructure updates.

Ultraviolet

April 2021 - June 2022

Unity Developer

- Led a small team of Unity developers, junior engineers, and contractors to build a metaverse prototype for the Octi social media app.
- Built a networked multiplayer prototype in Unity Mirror supporting 10+ concurrent players, with builds deployed to a Lightsail-hosted server and configured through Unity command-line parameters.
- Connected the live production user base to the Unity client by authenticating existing profile tokens against the company's REST API, then verifying each user server-side by UUID before granting access.
- Integrated the Unity stack into the studio's existing Docker deployment pipeline, previously Xcode-only, using unity-docker to containerize the correct Unity build.
- Increased community engagement by 20%, reaching #1 in the social media category for two weeks.

Evava VR

September 2020 - March 2021

Unity Developer (AR)

- Produced AR experiences using ARKit and Unity for venues including South Coast Plaza.

Konami Digital Entertainment

December 2019 - March 2020

Quality Assurance Tester

- Reported and tracked bugs with clear replication steps, and conducted console TRC compliance tests for several titles including *Yu-Gi-Oh! Legacy of the Duelist: Link Evolution* and *Skelattack*.

Tandon Online

June 2018 - August 2018

Unity Developer

- Networked all server-client interactions using Unity HLAPI, including player-specific blocks, crane mechanics, and user interface.

PROJECTS

WishSeekers: Solo Gameplay Programmer, Unity 6 / C#

Sole engineer on the full gameplay codebase of a 2.5D action roguelite in development. Built on a ScriptableObject data layer with all damage routed through one resolution pipeline (modifiers, crit, mitigation, absorb, afflictions) shared by 30+ abilities. Four NavMesh enemy archetypes and a multi-phase boss using animation-event hitboxes, telegraph windows and coroutine state machines. Game-feel stack spanning a melee combo state machine, i-frames, parry/counter and an input buffer with press-grace and release-edge windows, plus 9 custom URP/HLSL shaders, auto-registering object pools, Addressables scene streaming, custom editor tools and an Inspector-level balance guide for non-programmers.

Technologies: Unity 6/C#, URP/HLSL, ScriptableObjects, NavMesh, Addressables

Tower Arena: Third-Person Locomotion & Gunplay

Responsive third-person locomotion and gunplay built on animation state machines and blend trees, with IK for weapon handling and foot placement, procedural landing scaled to fall height, and a third-person camera reconciled with FPS-style aiming.

Technologies: Unity/C#, animation state machines, IK, HLSL

Firebot Auto Battler: Live-Service Systems, Twitch Integration

Twitch-integrated auto-battler running as a Firebot custom script. Persistent viewer accounts and weapon state across sessions, a currency system, and Twitch Prediction betting that resolves and pays out against battle results, with a real-time OBS overlay.

Technologies: Firebot, Twitch API (events, predictions), JavaScript, OBS overlay

LoL Draft Simulator: Supabase, Data Management, Full-stack Unity

Secure registration and login pipeline built on the Supabase C# SDK, with TypeScript automating real-time data manipulation. Python scripts scrape and import millions of player and game metrics via the CargoQuery API, feeding a scikit-learn regression model that scores roster strength.

Technologies: Supabase, TypeScript, PostgreSQL, Python, Unity/C#

SKILLS

Languages: C#, C++, Python, JavaScript, TypeScript, PostgreSQL, Java, OpenGL, HLSL

Gameplay: Animation state machines, blend trees, IK, procedural motion, combat systems, behavior trees, A* pathfinding, pooled-entity architecture, ScriptableObject data layers

Engines & Graphics: Unity 6, Unreal, SFML (C++), URP/HLSL shaders, Addressables, AR Foundation, ARKit

Networking: Unity Mirror, Unity HLAPI, client-server architecture, session state synchronization, REST API design, token-based authentication

Cloud & Build: AWS (API Gateway, Lambda, DynamoDB, Cognito, IAM, Secrets Manager, CloudFormation, Lightsail, CLI), Docker, unity-docker, CI/CD, Supabase

EDUCATION

B.S. Integrated Digital Media, Minor in Game Engineering

2015 - 2019

New York University, Tandon School of Engineering

Dean's List 2016-2017. Relevant coursework: Game Programming, Data Structures & Algorithms, Object Oriented Programming, Artificial Intelligence, Ideation & Prototyping.